

**VIETNAM GENERAL CONFEDERATION OF LABOUR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY**



NGUYEN QUANG HUY

**ASPECT-BASED SENTIMENT
ANALYSIS SYSTEM**

**INDUSTRIAL EXPERIENCE REPORT
COMPUTER SCIENCE MAJOR**

HO CHI MINH CITY, YEAR 2026

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Advised by
Assoc. Prof. Dr. Le Anh Cuong

HO CHI MINH CITY, YEAR 2026

ACKNOWLEDGMENT

I sincerely thank

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I hereby declare that this thesis was carried out by myself under the guidance and supervision of Assoc. Prof. Dr. Le Anh Cuong; and that the work and the results contained in it are original and have not been submitted anywhere for any previous purposes. The data and figures presented in this thesis are for analysis, comments, and evaluations from various resources by my own work and have been duly acknowledged in the reference part.

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ABSTRACT

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LIST OF ABBREVIATIONS

LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
GRU	Gated Recurrent Unit
NLP	Natural Language Processing
NLU	Natural Language Understanding
NLG	Natural Language Generation
HMM	Hidden Markov Model
CRF	Conditional Random Field
Viterbi	Viterbi Algorithm
BLEU	Bilingual Evaluation Understudy
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
METEOR	Metric for Evaluation of Translation with Explicit Ordering
CIDEr	Consensus-based Image Description Evaluation
SPICE	Semantic Propositional Image Caption Evaluation
DCG	Discounted Cumulative Gain
NDCG	Normalized Discounted Cumulative Gain
MRR	Mean Reciprocal Rank

Chapter 1

Introduction

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1.1 Context

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1.2 Motivation

1.3 Objectives

1.4 Contributing

Chapter 2

Related Work

2.1 Sentiment Analysis

2.2 Aspect-Based Sentiment Analysis

2.3 Stock Price Prediction from Sentiment Analysis

2.4 Research Gaps

Chapter 3

Methodology

3.1 Dataset

3.1.1 Data Collection

REFERENCES

Le, A.-C., Duong, T.-C., Nguyen, V.-H., and Le, T. V. (2025). Overview of the Legal-SLM shared task: Evaluating legal reasoning of Vietnamese small language models. In Mai, L. C., Huyen, N. T. M., and Trang, N. T. T., editors, *Proceedings of the 11th International Workshop on Vietnamese Language and Speech Processing*, pages 147–152, Hanoi, Vietnam. Association for Computational Linguistics.

APPENDIX